

Frequency

Emission code

Searchable

Accuracy

Range

VDF 118-137 MHz

A3E

line of sight propagation (space wave) [voice & codes]

A = ± 2 | B = ± 5 | C = ± 10 | D > 10

ADF-MDB 190-1750 MHz

NON-A1A
NON-A2A

vertically polarized signals
emphasizes phase

[voice relative bearing] $\pm 5^\circ$

200 nm $\rightarrow$ Day
70 nm $\rightarrow$ Night

(as desired)

Sense Aerial for $\Rightarrow 180^\circ$ ambiguity
Locator MDB $\rightarrow 10-25$ nm

Homing MDB $\rightarrow 50$ nm

Emergency MDB $\rightarrow > 50$ nm

VOR 108-118 MHz

AGW

Phase Comparison [voice & heading]
Fixed Aerial $\rightarrow 30$ Hz - 5 Hz

$\pm 5^\circ$

(omni)

Rotation $\rightarrow 30$ Hz - 1 Hz
D VOR $\Rightarrow$ Modulation change (all fixed aerodromes)

GLS 108-112 MHz

A8W

Will measure in depth of Modulation
Localizer $\rightarrow 180$ Hz $\rightarrow 90$ Hz [0.5 per dot]

Localizer $\Rightarrow 25$ nm $\rightarrow \pm 10^\circ$
17 nm $\rightarrow \pm 10^\circ - 35^\circ$
10 nm $\rightarrow \pm 35^\circ$

(dec. ~~180~~)
75 MHz

Right $\rightarrow 150$ Hz

(JRB)

Slide Slope $\rightarrow 4$ Hz $\Rightarrow 90$ Hz [0.14 per dot]

Slide Slope $\Rightarrow$ Max = 1.75 x θ
Min = 0.45 x θ

329-335 MHz

beam $\rightarrow 150$ Hz

(GS)

Max beam beam $\rightarrow$ height = 3000 ft

Minimum DH

OT Blue 2 Hz 400 Hz

Cat 1 200 ft > 200 ft

RNR req. > 550 m visibility > 800 m

OT Amber 4 Hz 1300 Hz

Cat 2 100 ft $100 > 30 > 200$

> 350 m

4 Hz White 6 Hz 3000 Hz

Yellow GS $\Rightarrow$ multiple of 3° Cat 3A nil < 100

> 200 m

GPS 1.1: 1575 MHz

(C/A code)

60 orbits $\rightarrow 20,200$ km $\rightarrow 55^\circ$ incline (12 hrs)

Vertical = ± 13 nm

95%

(P-code)

1.2: 1228 MHz

(P-code)

24 sat $\rightarrow$ 2.1 active, 3 spare (4 atomic clock/sat)
4 sat $\rightarrow$ 3D 5 sat $\rightarrow$ RAIM 6 sat $\rightarrow$ 4D + RAIM

Time = 40 msec

Receiver takes input of multiple combinations
1.5 m $\rightarrow$ max. clock error (3 sat / comb)

Best accuracy when 1 sat overhead
3 sat 120 apart

Frequency	Working	Mode A	Mode B	Mode C	Mode D	Mode S	Mode E
SSR 1030-1090MHz	Transponder responds with signal of 12 pulses 20.3 μ s sec apart, total 4096 codes Alert $\rightarrow$ for 20 sec. [SPI pulse after normal] Tones $\Rightarrow$ scrambling, twisting, Orth-interrogation	Mode A 8 μ s Identification	Mode B 17 μ s Identification Standing	Mode C 21 μ s Identification + P. Altitude	Mode D 28 μ s Tech. 100 ft	Mode S 16700000 codes eliminate errors	962-1213MHz PON Range 200nm ± 0.25 nm On lock on $\rightarrow$ 27 kbps VOR-DME $\Rightarrow$ in 600m
AWR 9375 MHz	Echo + Searchlight On Ground $\Rightarrow$ 110 ft + beam mode MAP mode $\rightarrow$ 150 ft + beam mode MAP mode $\rightarrow$ 150 ft + beam mode $\lambda = 3$ cm MAP mode $\rightarrow$ 150 ft + beam mode MAP mode $\rightarrow$ 150 ft + beam mode $\hookrightarrow$ weather + long range mapping	1. ROP high 50-2450 ft 2. ROP low 50-2450 ft 3. ROP low 50-2450 ft 4. ROP low 50-2450 ft 5. ROP low 50-2450 ft 6. ROP low 50-2450 ft 7. ROP low 50-2450 ft 8. ROP low 50-2450 ft 9. ROP low 50-2450 ft 10. ROP low 50-2450 ft 11. ROP low 50-2450 ft 12. ROP low 50-2450 ft 13. ROP low 50-2450 ft 14. ROP low 50-2450 ft 15. ROP low 50-2450 ft 16. ROP low 50-2450 ft 17. ROP low 50-2450 ft 18. ROP low 50-2450 ft 19. ROP low 50-2450 ft 20. ROP low 50-2450 ft 21. ROP low 50-2450 ft 22. ROP low 50-2450 ft 23. ROP low 50-2450 ft 24. ROP low 50-2450 ft 25. ROP low 50-2450 ft 26. ROP low 50-2450 ft 27. ROP low 50-2450 ft 28. ROP low 50-2450 ft 29. ROP low 50-2450 ft 30. ROP low 50-2450 ft 31. 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